PART A: Missing Value Analysis (Report)

Missing Value Summary Report

⇒The dataset was analyzed for missing values across all features. The percentage of missing data was calculated for each column to understand the extent of the issue.

⇒It was observed that the following columns contained missing values :-

- **Gender** had the highest missing percentage (~32.8%)
- **Region** had ~24.4% missing values
- **Age, bmi, cholesterol, and glucose** each had ~12% missing values
- **patient_id, blood_pressure, and disease_risk** had no missing values

⇒This indicates that both **categorical and numerical variables** are affected by missing data, which can impact model performance if not handled properly.

⇒To address this :-

- **Numerical features** were handled using various imputation techniques such as Mean, Median, KNN, and MICE.

- **Categorical features** were imputed using the Most Frequent (mode) strategy.
- Advanced techniques like **Random Sampling with Missing Indicator** were also applied to preserve data distribution.

⇒ Proper handling of missing values ensures :-

- No data loss
- Improved model accuracy
- Better statistical reliability

PART C: Final Dataset (Theory Explanation)

Final Data Pre-processing Report

After experimenting with multiple imputation and outlier handling techniques, the final dataset was prepared by selecting the most appropriate methods based on data characteristics.

1. Imputation Technique Chosen

The **MICE (Multiple Imputation by Chained Equations)** method was selected for handling missing values in numerical features (age, bmi, cholesterol, glucose).

- MICE considers relationships between variables
- Produces more realistic and statistically reliable values
- Performs better than simple methods like mean or median

For categorical variables (gender, region), the **Most Frequent (mode) imputation** method was used, as it preserves the most common category in the dataset.

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2. Outlier Treatment Method Chosen

The **Winsorization technique** was selected for handling outliers in numerical features.

- Caps extreme values instead of removing them
- Prevents data loss
- Maintains dataset size (important in healthcare data)
- Reduces the impact of extreme measurement errors

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3. Improvement in Data Quality

After pre-processing:

- All missing values were successfully handled
- Outliers were controlled without removing records
- The dataset became more stable and consistent

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4. Conclusion

The final dataset is now:

- Clean (no missing values)
- Robust (outliers handled effectively)
- Ready for machine learning modelling

These pre-processing steps improve model performance, accuracy, and reliability when predicting disease risk.